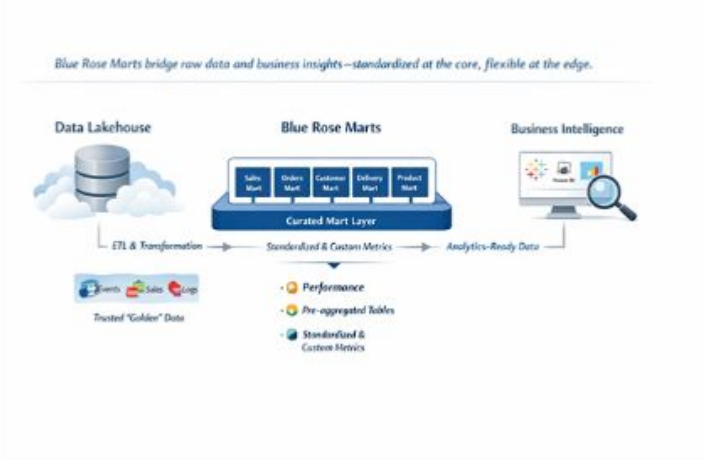
Olist E-Commerce Data Lakehouse - Rose Marts

Preview

From Golden Data to Business Impact: The Role of Blue Rose Marts



Note: Alt+Click navigate buttons below.

Why Rose Marts Matter	What Makes Up Rose Marts	How to Leverage - BI Demo

Why Rose Data Marts Matter?

From Trusted 'Golden' Data Lakehouse to Actionable, Business-Ready Insights — Powered by Blue Rose Marts

1. The Challenge (Before)

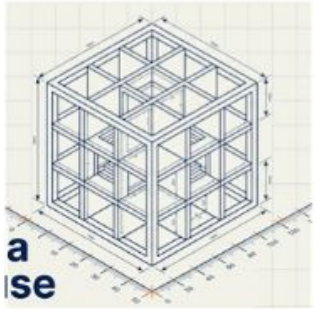
- **Data Silos**: Raw data is fragmented across sources and lacks consistency
- **High Latency**: Complex queries on raw data are slow and resource-intensive
- **Metric Drift**: Different teams report conflicting definitions of key metrics (e.g., "Revenue")

2. The Strategic Value of Blue Rose Marts

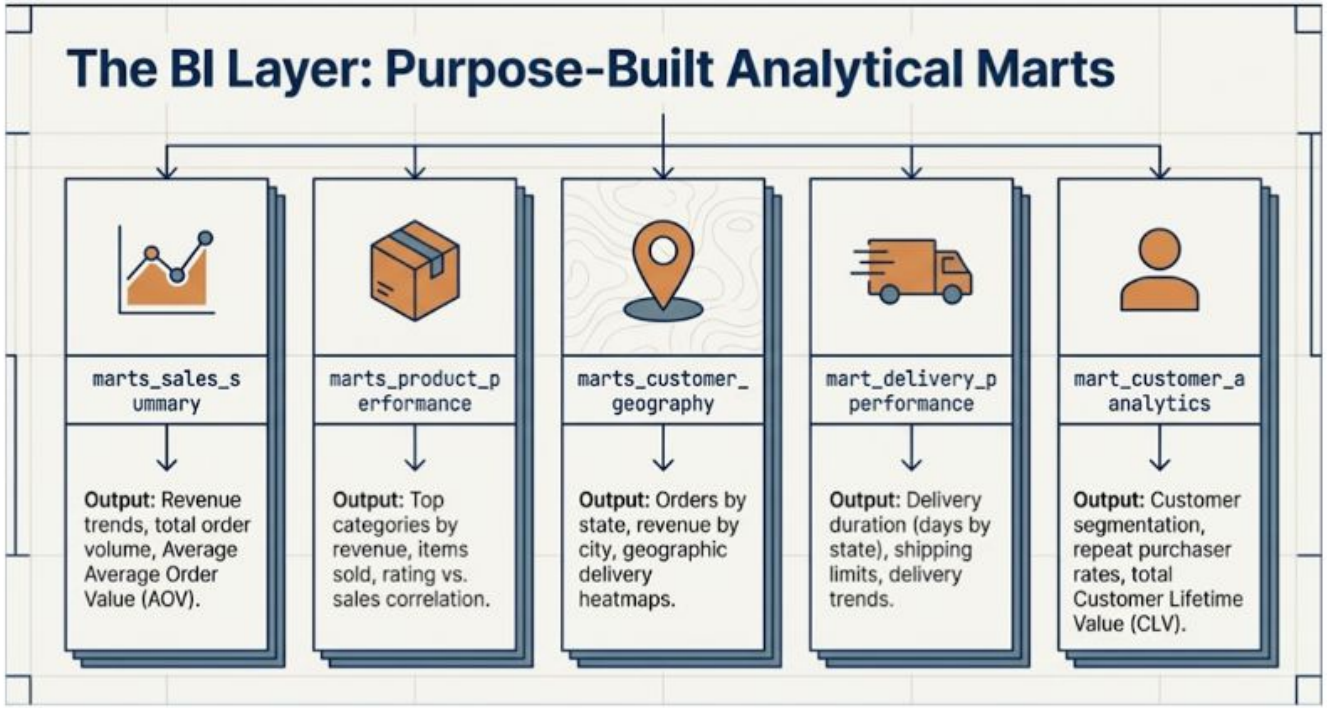
- **Performance**: Pre-aggregated, optimized tables significantly improve dashboard speed (up to 80% faster)
- **Governance**: Role-based access control ensures secure and controlled data usage
- **Consistency**: Standardized KPI definitions create a single source of truth across the organization

3. Key Outcomes — Blue Rose Marts

- **Cost Efficiency**: Reduced compute costs by querying smaller, well-modeled datasets instead of raw data
- **Self-Service BI**: Enables analysts to build reports independently without complex SQL joins
- **Reliability**: Ensures high data quality through automated testing in the transformation layer (e.g., dbt tests)



What Makes up Blue Rose Marts?



Blue Rose Marts represent the **business-ready layer** of the Olist data lakehouse, delivering **essential, standardized metrics** while enabling **flexible customization** to support diverse departmental needs.

How to Leverage Blue Rose Marts?

Executive Sales Overview *(Sample of 5,000 records from Jan 2017–Sept 2018)*

MoM revenue fell 4% in Aug 2018, despite total revenue rising 20% YoY. This demo illustrates the pipeline from Rose Marts to BI, enabling leadership to perform root-cause analysis.

Future enhancement:
city/state filter for drill-down insights.

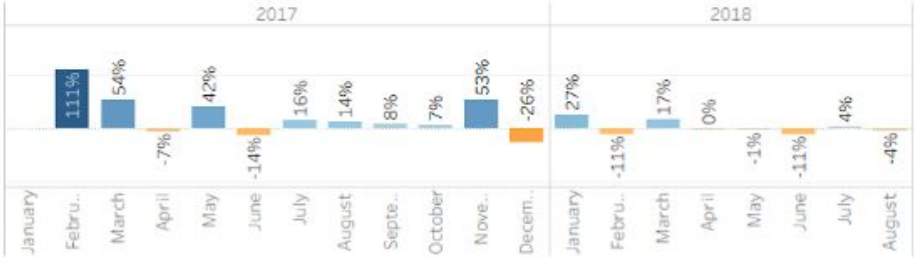
Total Revenue	Revenue MoM %	Total Order Volume	Avg. Order Value
$8.7M	▼ 4.1%	99K	$156
v.s yrs17 (20.1%)	(Aug v.s July 2018)		

Revenue Seasonality (Sales Heatmap)

Dark blue highlights peak sales months.

Revenue MoM Growth %:

Spiked in Nov 2017 (Black Friday), and then trended down to -4% by Aug 2018.



Average Order Value (AOV)

Steady around $156, dripped to $147 in Aug, then surged to $224 in Sept 2018.

Revenue by Day of Week (2018)

Higher spending observed on weekdays compared to weekends.